

contains some additional widgets as well. Most of the time, we can apply our working knowledge of Dialog to Xdialog, as it is.

The general form of an Xdialog command is as follows:

```
Xdialog <common options> <transient options> widget <widget options>
```

Here transient options are nothing but some Dialog options and other new options added to Xdialog.

Let's now dirty our hands with some Xdialog widgets, shall we? Issue the following commands:

```
Xdialog --backtitle "coloursel" --title "Color Selection Widget" \
--coloursel "Choose your favourite color" 0 6 123 234 85 &
```

```
Xdialog --backtitle "fontsel" --title "Font Selection Widget" \
--fontsel "Times 12" 0 6 &
```

```
Xdialog --backtitle "rangelow" --title "3 Range Selection Widget" \
--rangesbox "Select your values" 0 8 first 10 58 34 second 3 89 56 \
third 4 67 45 &
```

Figure 9 shows the corresponding Xdialog widgets. See how easily, with just some single-line commands, we are able to produce some true graphical widgets like colour selection, font selection, and range selection dialogue boxes?

(X)dialog with other languages

We can use Dialog and Xdialog from any programming language that supports calling external executables. The advantage lies in the fact that we have a way to add graphics to our programs without much effort. It could also serve as a way for rapidly prototyping GUIs.

Let's look at how to mix what we've learned, with Python scripts. Take a look at the following code snippet, where we create various widgets in a mixed manner. First it asks you the password key (that is, lfyrockz) and then after taking some more inputs, it provides a message to you in the end.

Please note that in the dictionary of Dialog and Xdialog commands, the various parameters are separated by a pipe character (|) to allow spaces in the various strings we pass. Copy the following code to a file (filename.py) and run it using python filename.py from a console:

```
#!/usr/bin/env python2.6

from subprocess import Popen, PIPE, STDOUT

# dictionary to hold various dialog and Xdialog commands.
dwdgtsatrb = {
    'dlgpswd': ('|dialog--stdout--insecure--backtitle|PasswordBoxWidget|'
    + '--title|passwordbox|--passwordbox|Enter the key|0|0|'),
    'xdlgname': 'Xdialog--title|XInputDialog|--inputbox|Enter your name|0|0|',
    'xdlgdob': 'Xdialog--title|XCalendar|--calendar|Select your DOB|0|0|',
    'dlgtime': ('|dialog--stdout--backtitle|TimeBoxWidget|'
    + '--title|timebox|--timebox|Select time of birth|0|0|',
    'msg' : 'Xdialog--title|XMsgBox|--msgbox|message|0|0|',
    )

def execute(scommand):
    """
    Routine to execute a command string and return (output, error, status)
    of the command.
    """
    l = scommand.split('|')
    p = Popen(l, stdout=PIPE, stderr=STDOUT)
    toe = p.communicate()

    return (toe[0], toe[1], p.returncode)

def main():
    """
    Application driver routine.
    """
    tpswd = execute(dwdgtsatrb['dlgpswd'])

    time = ()
    dob = ()
    ttime = ()

    if 'lfyrockz' == tpswd[0].strip():
        if not tpswd[2]:
            time = execute(dwdgtsatrb['xdlgtime'])

        if not time[2]:
            dob = execute(dwdgtsatrb['xdlgdob'])

        if not dob[2]:
            ttime = execute(dwdgtsatrb['xdlgtime'])

        if not ttime[2]:
            msg = 'Hi ' + time[0].strip() + '\n' +
            'Your DOB is : ' + dob[0].strip() + '\n' +
            'Time of birth is : ' + ttime[0].strip() + '\n' +
            'LFY wishes you the best.'
```



Figure 9: Widgets provided by Xdialog